

Intelligent Transportation Systems Make Use of Fog and Edge Computing for Navigation

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Abstract—The denizens of smart cities now enjoy improved access to navigational information through smartphones and other smart technologies. Addressing the challenges of data management, safeguarding sensitive information in both cloud and vehicular contexts, and elevating the dependability, efficiency, and utility of services become complex when dealing with extensive data volumes. This paper delves into the current state and future trajectory of the Internet of Things (IoT), particularly within intelligent transportation networks. The researchers propose that an automated edge/fog solution serves as a pivotal approach to resolving the dynamic ride-sharing problem. The shift of data processing and storage to the network's periphery is explored. The advent of fully operational next-generation 5G networks is anticipated to trigger a substantial increase in globally connected devices. Favorable conditions for cloud computing encompass low-latency, low-congestion networks with high user and sender density. The proliferation of Edge Computing (EC)-enabled networks augments portability, safety, and confidentiality. The term "EC architecture" encapsulates the design scheme of a system in this context.

Index Terms—Electric vehicles (EVs), Internet of Vehicles (IoV), Fog Computing (FC), Edge Computing (EC), Intelligent Transportation System

I. INTRODUCTION

In the contemporary era, the term "information boom" characterizes the current phase, presenting a pressing challenge in major cities. As the human population grows, our ability to explore the globe, share ideas, and acquire knowledge expands proportionally [1]. Urban dwellers in digitally connected cities increasingly utilize mobile phones, cars, and other devices for internet access. Technological advancements, including big data and cutting-edge technology, are integral to the urbanization process, giving rise to the concept of "smart cities" that enhance liveability and sustainability. These cities employ intelligent data management and sustainable development practices to improve municipal operations, such

as installing charging stations for electric vehicles [2]. With the proliferation of Internet of Things (IoT) devices, massive amounts of data are generated. To handle this data, IoT utilizes fog and cloud computing, making the data accessible through various channels. While data collected by IoT devices is stored in open data servers, analytics can be conducted locally in the fog layer, eliminating the need to send data to the cloud. The development of true "smart cities" necessitates innovative transportation solutions with precise and reliable IoT-based control mechanisms for real-time functionality [3]. Collaborative efforts by government agencies, enterprises, and non-profits, utilizing IoT and open data projects, enable the development of innovative services and applications for smart cities. Proper data management between customers and businesses is essential for these initiatives. Utilizing data from IoT devices, such as mobility and traffic flow data, allows for various applications, including micro- and macro-simulations, trip-time forecasts, and traffic flow visualization, contributing to the versatility and convenience of internet sharing [4]. Cloud storage and edge computing prove beneficial in minimizing the workload of managing and interpreting data from IoT devices, particularly regarding how vehicles are driven and how traffic moves. Approximation-based optimization techniques, such as heuristics, play a crucial role in efficiently addressing complex problems in Intelligent Transportation Systems (ITS) and similar systems [5]. Heuristics are especially vital for IoT analytics, considering the limited implementation of a hybrid cloud/edge architecture. The dynamic ride-sharing problem (DRSP) becomes a significant area of study due to its impact on modern cities, influenced by the fluid conditions typical of urban environments [6]. With the technological advancements, numerous IoT devices now communicate seamlessly. The installation of cellular base stations, edge routers, smart energy meters, and web-connected vehicles has further enhanced connectivity. The use of multiple data centers in

different physical locations is made possible through cloud computing. However, fog computing becomes crucial to prevent overwhelming these data centers with IoT device data, especially in scenarios where the cloud is distant from the user [7]. The term "fogging" is used to describe the challenge when the cloud is far from the user, impacting the speed of tasks over a large area. Fog computing, by positioning each fog node only one hop away from the user, addresses this challenge by enabling local processing without reliance on a remote cloud server. This proximity enhances communication between edge users and cloud data centers [8]. Fog computing solutions, equipped with memory and computational capabilities, enhance mobility, connectivity, computational capacity, distributed data processing, and cloud integration, particularly for applications with stringent latency requirements [9]. The proliferation of connected IoT devices serves as a key driver for big data, prompting the deployment of fog computing and fog networking to process the vast amounts of generated data [10]. Traditional network cores face challenges in managing the increasing demand for data transmission, leading to the development of Edge Computing (EC) as an alternative to traditional Cloud Computing (CC) infrastructures. Proximity to users' homes and workplaces enhances trust in the information by reducing latency and optimizing bandwidth utilization [11]. As the capacity of 5G wireless networks is strained by the deployment of overwhelming numbers of IoT devices, EC emerges as a solution to address the limitations of traditional CC infrastructures. Proximity, reduced latency, and efficient bandwidth utilization characterize EC, making it synonymous with cloud computing in discussions about localizing data and processing capacity [12].

II. CLOUD-BASED VEHICLE ROUTING USING THE INTERNET OF THINGS

The "internet of things" concept was pioneered due to the "things" worldview. A system connected to the internet can manage, monitor, and keep tabs on all these responsibilities [13]. In the not-too-distant future, every electronic device and the home appliance will be interconnected. The internet will serve as the connecting medium for all these devices. This novel concept has been equated to merging the digital and physical realms. That is the first word of a universal language shared by computers, mobile devices, tablets, servers, and more. It is the most effective method and has the backing of widely accepted standards of interaction. The remaining hardware, physical or purely digital, is collectively called "things." It has several different parts that can communicate with one another and share information [14]. Electronic components are integral to the operation of these gadgets. They can compute and interact with other networked devices. The distribution of essential utilities like water and electricity, the management of public transit, and the establishment of mobile health clinics are all examples of how cities may be made brighter. Widespread implementation of intelligent systems is essential for the success of any smart city [15]. Congestion in the city's transportation systems is a worldwide

issue. Management and control of traffic bottlenecks have significantly benefited from traffic signal control, intelligent parking systems, and intelligent route navigation. Using IoT, a "smart city" can be created. People living in a "smart city" know first-hand how crucial it is to regulate traffic. As the number of automobiles on the road increases, it becomes more challenging to locate a parking spot. Possible solutions include the development of innovative "smart" cities planned with this issue in mind. Many contemporary towns use networks of sensors to monitor traffic information. So, to sum up, what goes down: The usage of sensors could allow for better surveillance and upkeep of the parking lots. A new "smart" parking system will make finding a parking spot much more accessible. Travel times will be enhanced while congestion is decreased. They will show them the quickest route there, considering the time it will take and the likelihood of encountering traffic. This project aims to direct drivers to the most convenient parking garages and lots using Internet of Things technologies [16]. Transportation needs to be considered in any "internet of automobiles" research. Maps, places, pathways, and network topologies are only some of the many possible foundations for IoV routes. Those that are passionate about the subject of IoV have invested a lot of effort into researching it [17]. They investigated potential applications for IoV, its routing, and the availability of relevant simulation tools. When several cars are on the road, the system quickly adapts to the new configuration. This rapid rotation of networks is a significant reason viewer constantly tune in and out. The IoV could have many different unintended consequences. To drastically influence the functionality of the routing protocol, along with having high research value and various potential practical uses. They profoundly affect the protocol employed for transmitting data between networks. The likelihood of a car going from one location to another can be calculated after a model of its movement has been developed [18].

The prediction trail for vehicles over the next three days can be more detailed by including the interplay between cars and other components. Decisions on data transmission and forwarding are made based on the route the vehicle is projected to take. Intelligent traffic systems and the expansion of the Internet of Vehicles are designed with the comfort of drivers and passengers in mind. Despite appearances, a car's speed and direction might change for several reasons. It is also essential to know how the vehicle got there. Some people who should know better are utterly unaware of their collaboration. Users find a solution by combining explicit parameters with a probability matrix representing the paths taken by the cars and an association matrix representing their final destinations. The distance between vehicles is then determined by factoring in their planned routes [19]. The distance between cars and their respective caches plays a role in transferring information. Compared to the previous three, the new routing technique performs better in terms of packet delivery ratio, average end-to-end delay, and routing overhead

ratio. However, it still lags the top performer.

III. SAFETY-CENTRIC CLOUD ARCHITECTURE DESIGN

Users should prioritize cloud security across the whole development and planning process. As a rule, performance is prioritized over safety by cloud architects [20]. One of the most exciting benefits of cloud computing is the ability to store data online without the involvement of higher-ups.

A. Cloud-Based Security Architecture Is Crucial

As more businesses go to the cloud, users reap the benefits of scalability, productivity, efficiency, and cost savings. Since it allows for rapid decision-making and adaptation to market volatility, it could be helpful for businesses [21]. Organizations should avoid using cloud services due to the security risks they present. If there is not a solid cloud security framework in place, the chances of cloud computing could easily outweigh the benefits. As more and more of a company's processes are moved to the cloud, it becomes more difficult to guarantee data security. While much of it is housed in IT-approved systems, there is no doubt that much more hasn't been inspected. It can be challenging for businesses to keep tabs on information as it travels from internal servers to the cloud and other devices [22].

B. Cloud Security Knowledge

A trustworthy cloud computing system's three foundations are data security, integrity, and user access. A user's cloud infrastructure may be more secure if they are familiar with the parts shown in figure:1.

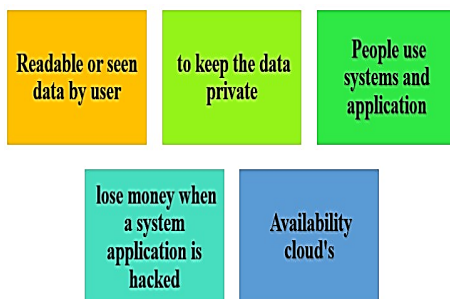


Fig. 1. Cloud Security Essentials

C. Preparing an Infrastructure as a Service (IaaS) Cloud for Safety

Infrastructure as a service provides various possibilities for keeping sensitive information safe. The API is the primary concern, while the other consideration is less important [23]. Even if the CSP also manages hardware, nothing will alter. Packet brokers and network nodes will handle the rest automatically (NPB). Figure 2 shows the primary components of the security architecture employed by IaaS cloud computing.

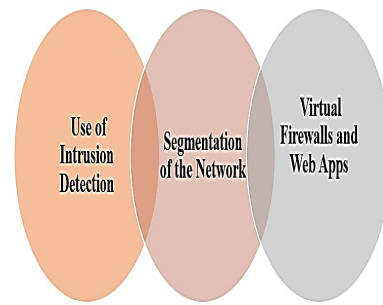


Fig. 2. IaaS Cloud Security Architecture

Infrastructure as a Service refers to online access to hardware, software, and networking components. This information is accessible to users worldwide. In this context, "cloud computing" refers to using an internet connection to contact a remote server [24]. Cloud providers are on the hook when it comes to protecting their hardware, software, data, and networks. It is also essential to prevent problems with the hypervisor, virtualization, and virtualization [25]. This business provides security for its customers by, among other things, avoiding the compromise of sensitive information and the malfunction of the software. The significance of these safeguards for IaaS cloud security architectures is highlighted in Figure 3.



Fig. 3. Infrastructure as a Service Cloud Security Models

D. Cloud Security Architecture Platform-as-a-Service

PaaS, which stands for "platform as a service," is a frequent term for the distribution and use of software across several computing platforms. However, this fails to account for the prohibitive resources and time needed to secure such programs. There is a heavy reliance on cloud security service providers to keep PaaS cloud infrastructure safe. The appropriate authorities will process the applications. Meanwhile, workers in the CSP sector must keep an eye out for any signs of misconduct among their peers. The following diagram highlights the fundamental distinctions between SaaS and PaaS cloud computing by comparing their respective security architectures [26].

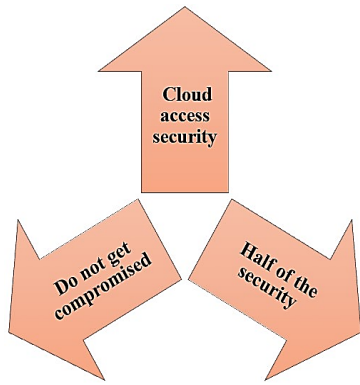


Fig. 4. The Security Features of the PaaS Cloud Computing Architecture

PaaS cloud service providers handle app security, whereas CSPs protect the underlying network and servers. One of the two PaaS options is built on top of an IaaS basis (IaaS). Solutions to these issues are depicted in Figure 4.

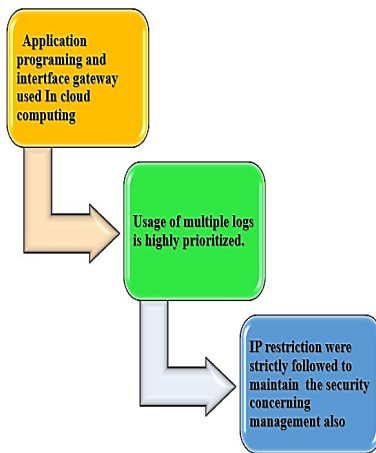


Fig. 5. Seven Security Measures for SaaS-Based Cloud Computing

According to the service agreement with the customer, the CSP is responsible for any security issues related to SaaS. SaaS may not care for a business's hardware, network, hypervisor, or operating system technicalities as shown in figure 6.

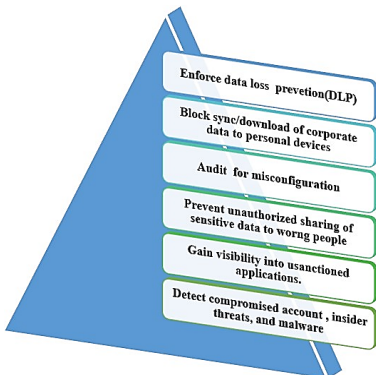


Fig. 6. Security Options for Software as a Service in the Cloud

IV. FUTURE VEHICLE SECURITY WITH EC-BASED ARCHITECTURE.

Cloud computing, with its combination of processing power and data storage, offers a versatile solution for vehicles . It enables remote file storage and collaboration, overcoming limitations in vehicle storage and processing capabilities. The efficiency of uploading and retrieving information from cloud servers is crucial in utilizing cloud computing effectively. Increased connectivity on roads facilitates quicker detection of app glitches and service disruptions, but it also intensifies data exchange challenges and power consumption for wireless devices [27]. The Cloud Cover, a pivotal aspect of cloud infrastructure, provides extensive storage and facilitates rapid data analysis . The Edge Cloud layer enhances real-time communication, allowing seamless coordination among transportation industry workers and efficient management of applications with low latency. Additionally, it supports video analytics and location-based applications .The Intelligence for Motor Vehicles layer envisions enhanced cooperation among vehicles through information sharing, service negotiation, and goods storage [28]. In this intelligent vehicular layer, vehicles utilize built-in sensors, GPS, cameras, and other equipment to transmit data for interpretation and coordination. "Smart vehicular computing" enables collaborative efforts among vehicles on the road .SDN-Enabled Parts, a crucial element for Vehicular Edge Computing (VEC), addresses the challenge of managing numerous vehicles and periodic connection failures. VEC implementation enhances the vehicle network, paving the way for new automotive applications and emphasizing the importance of balancing communication and processing needs for future advancements.

V. FUTURE PATHS FOR VEHICLES ON THE PERIPHERY

The advancement of Edge Computing (EC) lags behind academic and commercial systems despite substantial progress and the removal of various obstacles. When designing for scalability and portability, crucial factors such as security, privacy, device heterogeneity, computation offloading, reliability, context awareness, and caching must be considered. The imminent speed enhancements in 5G networks will facilitate more complex algorithms, offering improved performance; however, the deployment of 5G networks in remote areas poses significant challenges [29]. Efficient utilization of time and power involves offloading work from end devices to edge nodes, optimizing resource use like memory and power. The impact of 5G on edge computing and wireless connections remains uncertain in the European Union. Potential improvements include increased bandwidth, decreased latency, and heightened security challenges for IoT devices [30]. As the number of mobile devices connecting to the internet via the periphery increases, adaptability improves but may lead to service quality issues. Fault-tolerant systems can ease transitions between networks and administrative domains. Security requirements vary depending on hardware and software. Privacy concerns intensify with the transmission of sensitive information, requiring compatible security techniques in EC

designs. Cooperation with devices of varying network connection expectations is essential for seamless network improvements. App-aware network and device management enhance communication and resource utilization at the network edge. Ensuring the edge system functions independently of cloud servers is crucial. Challenges include power consumption, network latency, device availability, and portability. To enforce storage policies across diverse edge devices, cloud system administrators must address challenges in data format, storage, API, and OS decisions. Tailored data models for each layer enhance the current setup. Businesses can now provide transparent pricing for cloud-based payment solutions, although valuing Edge Computing is more intricate. Telecom companies use user information for billing rates, suggesting potential pricing flexibility for the service.

VI. CONCLUSION

"Fog computing" is the term used to describe the practise of relocating computation to locations closer to end consumers. While several solutions exist for storing and retrieving data in the cloud, this one takes a slightly different approach. Now that we have cloud computing, we can respond swiftly. Users using this technology boost the network's effectiveness by decreasing congestion. Calculating different types of fog. Using cloud computing, processing and calculating times can be reduced. The challenges of cloud computing are explored in depth in this essay. The concept of "fog computing" is likewise touted as a cutting-edge innovation. As the number of Internet of Things (IoT) gadgets continues to rise at an exponential rate, fog computing has emerged as the most effective solution. Fog computing allows tasks to be completed much more quickly and independently of a network, and much closer to the end user. That is, roughly speaking, the point being made. The technologies that make up fog computing are studied. Conventional forms of transportation may become outdated due to edge, cloud, and fog computing. The growth of open data initiatives in smart cities increases the likelihood that residents of these communities will become aware of such discrepancies. Users of both public and private transportation services can benefit from these freely available data sets, which provide information that is both current and accurate. This information has the potential to enhance customer journey planning through the application of machine learning and optimization techniques. An increasing number of transit agencies are installing bus tracking systems for practical, economical, and customer service reasons.

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